# Round 7 – Computing the Three Per-Neuron Corrections Cheaply

Research analysis

## Executive conclusion

The supplied measurements say the remaining error is concentrated in only three marginal quantities per neuron: variance, diagonal $\kappa_3$, and diagonal $\kappa_4$. The channel oracle gives final-MSE ratios 0.533, 0.858, and 0.699 when each channel is made exact, so variance is the largest measured opportunity.

The strongest cheap result is an exact identity for all requested target variances: if $W = [w_1, \ldots, w_n]$ and $C$ is the post-ReLU covariance, then

$$\mathbf{v} = \operatorname{diag}(W^\top C W)$$

can be evaluated exactly with one dense $n \times n$ by $n \times n$ product, $CW$, followed by $n$ dot products. At $n = 1024$ this is about 1.001 u per layer. It is not cheaper than the existing full-covariance propagation by itself, but it is a clean oracle and a useful implementation baseline.

For further savings, the most defensible experiment is not top-$k$ weight truncation (cross terms make that approximation structurally unsafe), but a *target-neuron selective correction*: preserve the chain's variance and estimate only the residual on the 300–400 important neurons using stored exact $C$ and a low-rank/diagonal decomposition of the *error covariance*. Its feasibility can be decided directly from one stored layer.

For $\kappa_4$, the previous deep-research draft contained a dangerous shortcut: the $(2, 1, 1)$ fourth-cumulant term is a fourth-order object, not a quadratic form of the third-cumulant slice. The correct contraction requires a $K_4(j, j, :, :)$ family. The useful new route is therefore to approximate the *birth contribution* to these $K_4(j, j, :, :)$ slices from the local ReLU closure, then contract the resulting structured terms with each target row $w_i$ without storing the full four-tensor.

## Q1. Variance channel

### Q1.1 Exact target-only evaluation

**Derived.** For target neuron $i$,

$$v_i = w_i^\top C_{\text{post}} w_i.$$

Let $m$ be the number of neurons with $|\mu_i/\sigma_i| < 0.75$ (roughly 300–400). Form only the selected column matrix $W_T \in \mathbb{R}^{n \times m}$. Then

$$V_T = \operatorname{diag}(W_T^\top C_{\text{post}} W_T).$$

One multiplication $C_{\text{post}} W_T$ costs

$$nm(2n - 1) \approx 2n^2 m$$

FLOPs, i.e.

$$\frac{m}{n}\,\text{u} \qquad \text{in the } n^3 \text{ cost unit.}$$

For $m = 350$ this is 0.342 u, plus about $m/n^3 \approx 3.3 \times 10^{-7}$ u for the final diagonal dot products.

**State.** Full $C_{\text{post}}$ (already present), plus the selected $W_T$. No new persistent tensor.

**Important distinction.** This is an exact way to compute the 300–400 requested variances, and it is about one-third of a dense $n^3$ product, but it does not by itself remove the cost of generating the accurate $C_{\text{post}}$. Thus its end-to-end value depends on whether the covariance update can also be made selective.

### Q1.2 Selective residual covariance correction

**Conjecture.** Write

$$C_{\text{post}} = C_0 + E,$$

where $C_0$ is the chain value and $E$ is the off-diagonal error. The target correction is

$$\delta v_i = w_i^\top E w_i.$$

The measurements say about 95% of the variance error is this random projection of the previous layer's off-diagonal covariance error, with a fresh component around $4.2 \times 10^{-5}$ RMS per entry.

A new test is to estimate whether $E$ is low-rank *as seen through the target rows*, even though $C_{\text{post}}$ itself is not low-rank. Take the exact stored difference

$$E = C_{\text{true}} - C_{\text{chain}}$$

and compute its randomized SVD. If

$$E_k = U_k \Sigma_k V_k^\top,$$

then

$$\delta v_i \approx \sum_{a=1}^{k} \sigma_a (w_i^\top u_a)(w_i^\top v_a).$$

For symmetric $E$, this simplifies to $\sum_a \lambda_a (w_i^\top u_a)^2$.

**Cost.** Once the $k$ eigenvectors are available, evaluating all $m$ target corrections costs about $O(nmk)$ FLOPs, negligible here. The expensive part is obtaining $U_k$ from stored $E$; a randomized rank-$k$ decomposition using $q$ power/orthogonalization passes costs roughly $(2q+1)k$ u for full $n \times n$ multiplies. For $k = 16$, $q = 1$, this is about 48 u as a one-time diagnostic, not an every-layer production cost.

**Falsifier.** On one stored layer, measure the exact target correction vector $\delta v$ and compare it with the rank-$k$ projection for $k = 4, 8, 16, 32$. If $k = 32$ still exceeds 0.1% relative RMS on the 300–400 target neurons, reject this route. This test is comfortably under one hour because it uses only $1024 \times 1024$ NumPy matrix operations.

### Q1.3 Why top-$k$ weights should be rejected

**Reject.** The tempting approximation

$$w_i^\top C w_i \approx \sum_{j,k \in T_i} w_{ij} w_{ik} C_{jk}$$

truncates both diagonal and cross terms. Because the measured residual is approximately random across off-diagonal entries, there is no reason for the covariance error to concentrate on the largest-magnitude entries of $w_i$. Therefore the apparent $O(mk^2)$ cost is misleading: the omitted cross terms can dominate the correction.

**First falsifier.** This is already testable from stored arrays: rank neurons by $|w_{ij}|$, evaluate the partial quadratic form for $k = 8, 16, 32, 64$, and compare with the exact quadratic form. Require the residual relative error to be below 0.1%. Failure at $k \leq 64$ kills the method immediately.

### Q1.4 Control-variate sampling of variance corrections

**Conjecture.** If one samples the correction

$$\delta v_i = \mathrm{Var}(z_i')_{\mathrm{MC}} - v_i^{\mathrm{chain}},$$

the chain prediction removes the dominant deterministic component. But the measured fresh covariance error is $\sim 4.2 \times 10^{-5}$ per entry, so the remaining quadratic-form variance is not known to be small enough for $10^4$–$10^5$ samples.

A 16-layer sample is quoted at 0.016 u. Thus $S = 10^4$ and $10^5$ samples cost about 160 u and 1600 u, respectively. Only an observed variance reduction of roughly 100x or more could fit the leaderboard budget.

**Falsifier.** With one network, run $S = 2000$ antithetic samples and directly measure the sample variance of $\delta v_i$ on the target neurons. Extrapolate the sample count required for 0.1% relative RMS. If it exceeds about $2 \times 10^4$ samples, reject as a production method.

### Q1 end-to-end

Replace the chain variance on the target neurons with either the exact selective quadratic form or the validated residual correction. Rerun the 16-layer mean propagation. The channel oracle says the theoretical ceiling is a final-MSE reduction from $2.29 \times 10^{-8}$ toward $1.22 \times 10^{-8}$, before any interaction with the other two channels.

## Q2. Deep $\kappa_4$ diagonal

### Q2.1 Correct tensor structure

**Derived.** For target row $w_i$, the fourth cumulant is

$$\kappa_4(z_i') = \sum_{a,b,c,d} w_{ia} w_{ib} w_{ic} w_{id}\, K^{\mathrm{post}}_{4,abcd}.$$

The $(2,1,1)$ sector is

$$T_i^{211} = 6 \sum_j w_{ij}^2 \sum_{k \neq l} w_{ik} w_{il}\, K^{\mathrm{post}}_{4,jjkl}.$$

Define the slice family

$$M_j[k,l] = K^{\mathrm{post}}_{4,jjkl}.$$

Then

$$T_i^{211} = 6 \sum_j w_{ij}^2\, w_i^\top M_j w_i.$$

This is the key structural reduction: the expensive object is a family of $n$ matrices, not a generic rank-4 tensor.

The $(1,1,1,1)$ sector can be written similarly as a residual four-index contraction after removing repeated-index sectors. Without an additional closure, it cannot be reduced to the stored $K_4(i,i,j,j)$ slice alone.

### Q2.2 New cheap route: local-birth factorization

**Conjecture.** The measured fact that the deep $K_4$ diagonal is regenerated at every ReLU suggests treating the new $(2,1,1)$ content as a local birth term. Under a second-order multivariate Edgeworth/ReLU closure, the leading source is a sum of products of local derivatives with the pre-activation $\kappa_3$ and covariance:

$$M_j \approx \alpha_j \operatorname{diag}(a_j) C \operatorname{diag}(b_j) + \beta_j (C \circ A_j) + \gamma_j R_j,$$

where $A_j$ is a vector derived from the stored $K_3(j,j,:)$ slice, and the scalar coefficients $\alpha_j, \beta_j, \gamma_j$ are exactly the ReLU/Hermite coefficients already used by the chain. The important property is separability: each birth term is a sum of rank-1 or Hadamard-times-covariance pieces, so its contraction with $w_i$ reduces to dot products involving

$$Cw_i, \qquad (A_j \circ C)w_i,$$

without materializing a four-tensor.

**Cost.** For all target neurons, precompute $CW_T$ at about $m/n$ u $\approx 0.34$ u. For $k$ separable birth terms, the additional cost is approximately $km/n$ u for the needed projected covariance vectors plus $O(knm)$ elementwise work. For $k = 4$, this remains below about 2 u per layer.

**Falsifier.** Use stored true $K_4(i,i,j,j)$ and $K_4(i,i,i,j)$ plus the captured chain quantities to fit the coefficients $\alpha, \beta, \gamma$ at one layer. Then predict the target-vector $\kappa_4$ correction for the next layer. Reject if the relative RMS error on neurons with $|\mu/\sigma| < 1.5$ remains above 10% or if the fit is unstable across layers 8–16.

### Q2.3 $(1,1,1,1)$ term

**Measure-this, not assume.** The all-distinct sector is the least identifiable from the stored arrays. A useful diagnostic is a randomized contraction oracle. Draw $q$ random target vectors $r_s \in \mathbb{R}^n$ and estimate

$$y_s = \sum_{a,b,c,d} r_{sa} r_{sb} r_{sc} r_{sd} K^{\text{post}}_{4,abcd}$$

from the teacher arrays, then ask whether the residual after the local-birth 211 model is low-dimensional. If the residual requires a large number of probes, a cheap 1111 representation is implausible.

A production tensor-network construction should therefore be attempted only if this diagnostic finds a numerical rank comfortably below about 32.

## Q3. Sampling the three corrections

**Derived budget constraint.** At the supplied sample price, $10^4$ full-network samples cost roughly 160 u and $10^5$ cost roughly 1600 u. The current chain is 260 u. Therefore raw sampling cannot be the main estimator unless the effective variance is reduced by about two orders of magnitude.

**Conjecture.** Antithetic sampling is likely to help little for these particular corrections because the target network is positively homogeneous but ReLU destroys the odd/even cancellation that would be needed for very strong variance reduction. Quasi-random sphere designs may improve constants, but the supplied measurements already indicate that the problematic covariance and $K_4$ patterns are high-dimensional and random.

**Best sampling test.** Use the chain's own Gaussian surrogate as the control variate and estimate only

$$\delta_s = Y_s - Y_s^{\text{chain}}.$$

Run $S = 2000$ antithetic pairs. Estimate the variance reduction factor for each of the three per-neuron corrections. A 100x reduction is the threshold that would make $10^5$-sample-class estimators competitive with the current 260-unit analytic chain.

## Q4. What to run first

The ranking below is based on the supplied measured channel oracles and the incremental cost of the proposed experiments, not on a claim about leaderboard methods.

| Channel | Measured opportunity | Diagnostic cost | Priority |
|---|---|---|---|
| Variance | 47% MSE reduction | 0.34 u target quadratic form | 1 |
| $\kappa_4$ diag. | 30% MSE reduction | $\sim$ 1–2 u local-birth model | 2 |
| $\kappa_3$ diag. | 14% MSE reduction | tens of units if sampled | 3 |

The first experiment should therefore be the stored-array test of the variance correction subspace:

$$E = C_{\text{true}} - C_{\text{chain}}, \qquad \delta v = \text{diag}(W_T^\top E W_T).$$

Measure how many eigen-directions of $E$ are needed to reproduce $\delta v$ to 0.1% on the 300–400 relevant neurons. This directly answers whether the random $4.2 \times 10^{-5}$ per-entry covariance error is nevertheless low-dimensional *after projection onto the only quadratic forms that matter.*

If that test fails, move immediately to the Q2 local-birth factorization test. Only after both analytic routes fail should the sampling experiment be run.

## End-to-end acceptance criteria

1. Variance: target-neuron relative error $\leq 0.1\%$ and a measurable final-MSE reduction toward the $1.22 \times 10^{-8}$ oracle.
2. $\kappa_4$: relative error $\leq 10\%$ on the closure-relevant neurons, with a final-MSE reduction toward the $1.60 \times 10^{-8}$ oracle.
3. Production budget: any combined method should remain well below the 1024-unit hard budget, and preferably close to the leaders' 133–165-unit range.

## Bottom line

The strongest new fact is that the variance problem can be made *exact for the important 300–400 neurons* at only about 0.34 u of matrix-multiply work per layer once $C_{\text{post}}$ exists. The remaining research question is whether the covariance *error* itself has a low-dimensional image under these target quadratic forms. That is a falsifiable stored-array question.

For $\kappa_4$, the correct target is not a third-cumulant quadratic form. The viable route is a local-birth factorization of the $(2, 1, 1)$ and, if necessary, residual $(1, 1, 1, 1)$ sectors. Its success or failure can be determined from the already-stored slice arrays before any expensive production implementation.